An SRAM-Based Digital Compute-in-Memory Macro with Dual-Bit Input Data Sparsification and Restructuring for Energy-Efficient MAC

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***Abstract*—SRAM-based digital computing-in-memory (CIM) provides a low-power, high-precision method for multiply-and-accumulate (MAC) operations. However, input data with a high toggle rate and low sparsity will significantly increase the energy consumption of digital CIMs. This work presents an input-transforming and weight-precomputation (ITWP) SRAM-based digital CIM macro. Two strategies involving input sparsification and input restructuring are proposed to transform the dual-bit input data, resulting in higher sparsity and lower toggle rate. The sum of the weights is precomputed to correct the partial sum (PSUM) of the sparsified input data. A 13T-based hybrid SRAM CIM unit (HSCU) and a split adder tree (SAT) are proposed to compute on the restructured two input sets with different sparsity levels, leading to a higher throughput and energy efficiency. Finally, the ITWP CIM macro is implemented in a 28 nm process and achieves a peak throughput of 0.935 TOPS and an energy efficiency of 20.08–33.78 TOPS/W with signed 8b input and weight. Compared to the conventional bit-serial CIM macro, the ITWP CIM macro is demonstrated to increase the energy efficiency to 1.42x and up to 3.04x.**

***Index Terms*—SRAM, compute-in-memory, input transforming, weight precomputation, split adder tree**

I. Introduction

Artificial intelligence (AI) edge devices offer notable advantages for AI tasks, including enhanced privacy, reduced latency, and increased bandwidth, but are often limited by power constraints [1], [2]. Unlike traditional von Neumann-based systems, CIM architectures significantly increase the throughput and energy efficiency of MAC operations, making them a superior choice for AI edge devices.

SRAM-based CIM has received more attention due to its high speed and high reliability [3]–[5]. The analog CIMs have advantages in terms of routing and energy efficiency. Nevertheless, PVT (process, voltage, temperature) and signal margin have a significant impact on the accuracy of analog CIMs [4]–[7]. Digital CIMs, on the other hand, are more accurate and robust [8]–[10]. However, digital CIMs confront a trio of significant challenges: (1) Digital CIMs experience a significant increase in energy consumption as the input toggle rate increases. (2) A massive adder tree used to sum the partial products, leads to a considerable energy consumption. (3) The computational parallelism of digital CIMs is limited by routing congestion, resulting in low throughput.

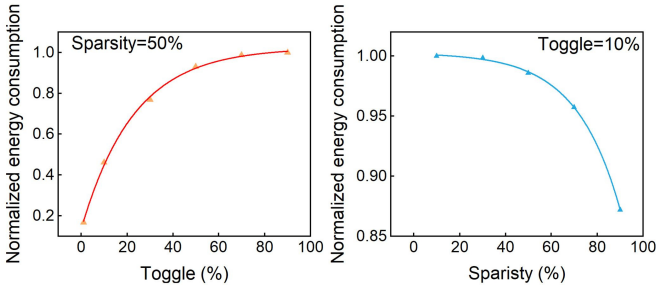


Fig. 1. (a) Normalized energy consumption of digital CIMs with different input toggle rates at 50% input sparsity. (b) Normalized energy consumption of digital CIMs with different input sparsity levels at 10% input toggle rate.

Considerable effort has been devoted to improving the energy efficiency of digital CIMs, specifically targeting optimizations in the adder tree and the local computing unit. For example, Chih et al. improved energy efficiency by 30% using a 14/28T hybrid adder tree [9]. Wang et al. considerably boosted energy efficiency by adopting bit-wise OR operations to replace addition with no overlap partial products [11]. Guo et al. reduced the number of adders by approximating the partial products [12].

In addition, transforming the properties of the input data can significantly improve the energy efficiency of digital CIMs. For example, minimizing the toggle rate significantly reduces energy consumption, as shown in Fig. 1(a). Similarly, improving the sparsity of the input data can further reduce energy consumption, as shown in Fig. 1(b). However, the exploitation of these features remains underutilized in current research.

This work proposes an input-transforming and weight-precomputation (ITWP) CIM macro to perform dual-bit MAC operations with markedly enhanced energy efficiency. The primary contributions are as follows: (1) An input sparsification strategy is proposed in which the low-sparsity input data is inverted, resulting in higher sparsity and a lower toggle rate. The precomputed and stored sum of the weights is employed to correct the result with minimal overhead. (2) An input restructuring strategy is introduced, designed to restructure the sparsified dual-bit input data. This strategy notably reduces the toggle rate, resulting in lower energy consumption in MAC operations. (3) To align with the proposed data flow

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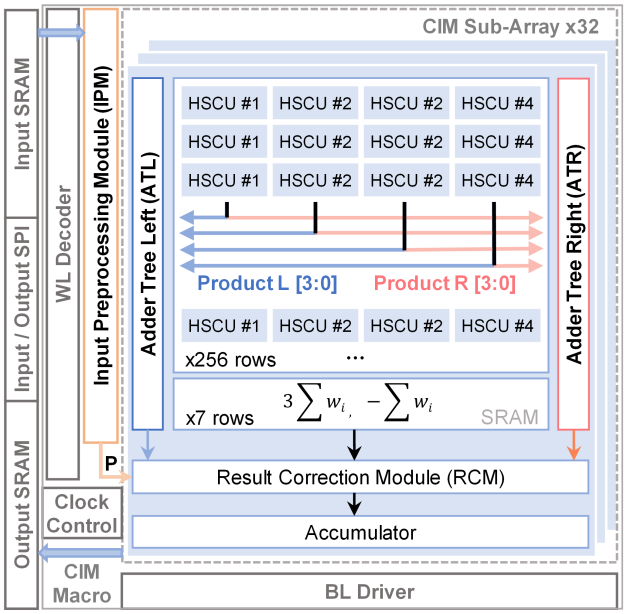


Fig. 2. The overall architecture of the proposed ITWP CIM macro.

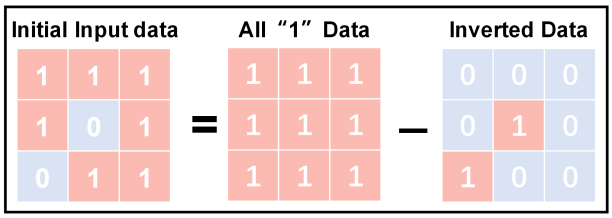


Fig. 3. Input sparsification principle.

processing method, a customized CIM macro architecture and key circuit modules, including a 13T-based hybrid SRAM CIM unit (HSCU) and a split adder tree (SAT), are proposed to simultaneously process the transformed dual-bit input data, resulting in higher throughput and energy efficiency. With these technologies, the ITWP CIM macro achieves a 2x throughput, a 1.42x energy efficiency, and up to a 3.04x efficiency under specific input data compared to the conventional bit-serial CIM macro.

II. PROPOSED ARCHITECTURE

A. Overall Architecture of the ITWP CIM Macro

Fig. 2 shows the ITWP CIM macro architecture. This macro contains 32 CIM sub-arrays. Two sub-adders, an accumulator, a result correction module (RCM), 256x4 HSCUs, and 7x4 SRAM cells make up each sub-array of the ITWP CIM macro. The input preprocessing module (IPM) transforms the input data and generates the control signal P. The sub-adders sum the partial products generated by the restructured two input sets. Note that the sub-adders are placed on either side of the SRAM array, with staggered routing to reduce routing congestion. The RCM corrects errors in MAC results, which could happen due to handling the input data's sign bit (MSB) or inverted input data. The 7-row SRAM cells store the precomputed sum of the weights. The accumulator sums up the results over multiple cycles. The BL driver and the WL decoder control the writing and reading of the weights.

TABLE I. INPUT RESTRUCTURING PRINCIPLE

\	Dual-bit input		Unsigned bit			Sign bit		
\	IN[i]	IN[i+1]	INR	INL	SHIFT	INR	INL	SHIFT
Place value	1	2/-2	1	2	\	-1	2	\
\	1	1	1	1	0	1	0	0
\	1	0	1	0	0	1	1	0
\	0	1	1	0	1	1	0	1
\	0	0	0	0	0	0	0	0

TABLE II. RESTRUCTURED TOGGLE RATE

Sparsity*(%)	Toggle(%)	INL Toggle(%)	INR Toggle(%)	Avg. Toggle(%)
90	18.61	1.75	31.13	16.44
80	32.07	7.27	46.17	26.72
70	42.57	16.07	50.86	33.47
60	47.81	26.47	48.90	37.69
50	50.05	37.16	37.24	37.20

* Input data with less than 50% sparsity is inverted.

B. Input-Transforming Principle and Proposed Circuit

The ITWP CIM macro employs two distinct strategies to transform input data, thereby markedly enhancing the energy efficiency of MAC operations.

Input sparsification: As shown in Fig. 3, the difference between a set of all "1" data and the bit-wise inverted input data can be used to represent the initial input data equivalently. Besides, the MAC result for a set of all "1" data and weights corresponds exactly to the sum of the weights, which is precomputed and stored for ready access. Therefore, when the input data with a sparsity of less than 50%, it can be sparsified by inverting. Finally, the PSUM can be subtracted from the sum of the weights to get an accurate PSUM.

Input restructuring: In each cycle, 256 dual-bit input data are fed into the macro. After sparsification, for the unsigned input data, IN[i] and IN[i+1] are restructured into input left (INL) and input right (INR), where INL = IN[i] & IN[i+1], and INR = IN[i] | IN[i+1], as shown in Table I. The average toggle rate of input data with different sparsity levels decreases after restructuring, as shown in Table II. Notably, as depicted in Fig. 1(a), the impact of a reduced toggle rate on energy consumption outweighs that of an increased toggle rate. Consequently, processing the restructured input data can significantly reduce energy consumption. In addition, when IN[i] = 0 and IN[i+1] = 1, the partial products of INR should be shifted, denoted by SHIFT = 1. For the sign bit of the input data, simply change INL by encoding INL = IN[i] & $\overline{\text{IN[i+1]}}$, which ensures a high encoding similarity between the sign bit and the unsigned bit.

Fig. 4 shows the detailed circuit of the IPM. Firstly, 32 input data are used for the input sparsity detector (ISD) to detect if the sparsity is below 50%. As the number of 1s in the input data increases, the propagation delay for the continuously inverted signal DEN decreases. The sparsity of the input data in the reference delay chain is 50% (REF_0, REF_1, REF_2 ... REF_{31} = 1, 0, 1...0), which serves as a benchmark. As shown in the operating waveforms, when the partial input data's sparsity is higher than 50%, $DREF_{31}$ is pulled high earlier than $DOUT_{31}$, resulting in P_0 to be stuck at 0. Conversely, P_0 is pulled high and PB_0 is stuck to 0. The enable signal of the output latch is (P_0 | PB_0) & DEN, ensuring P is not changed until the next cycle. Subsequently,

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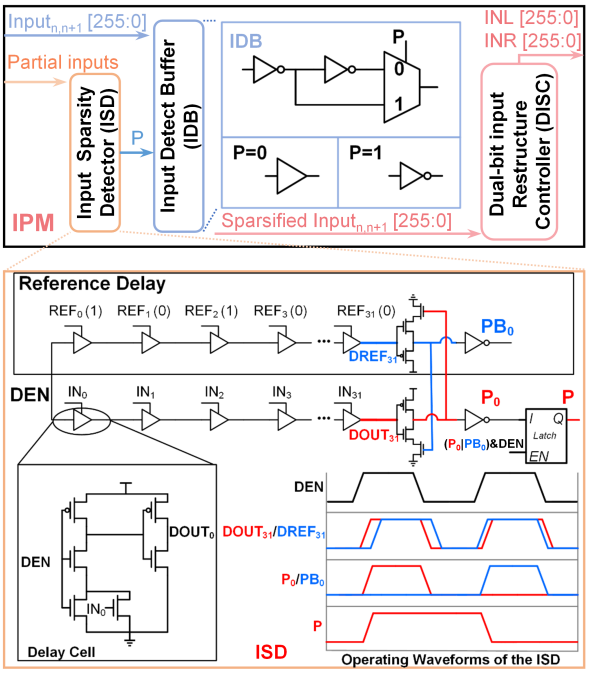


Fig. 4. The architecture and circuit of the IPM.

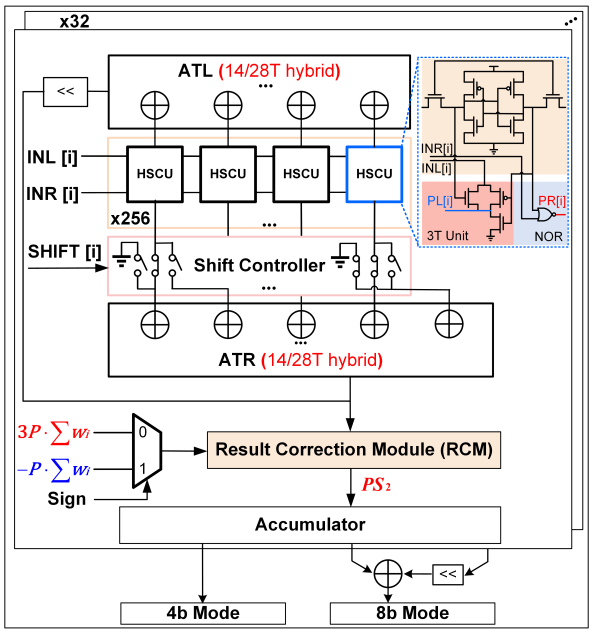


Fig. 5. The split adder tree and the hybrid SRAM CIM unit.

the input detect buffer (IDB) inverts the input data when P is 1, leaving it uninvited otherwise. The output from the IDB is then restructured by the Dual-bit input restructure controller (DISC). In the next cycle, the INL and INR are fed into the SRAM CIM array.

C. Split Adder Tree and Hybrid SRAM CIM Unit

As shown in Fig. 5, the adder tree is divided into two sub-adder trees, the adder tree left (ATL) and the adder tree right (ATR), which are interleaved using 28T and 14T full adders. The partial products derived from INL are summed by the ATL, while those derived from INR are summed by the ATR. Since INL and INR have different sparsity levels and toggle

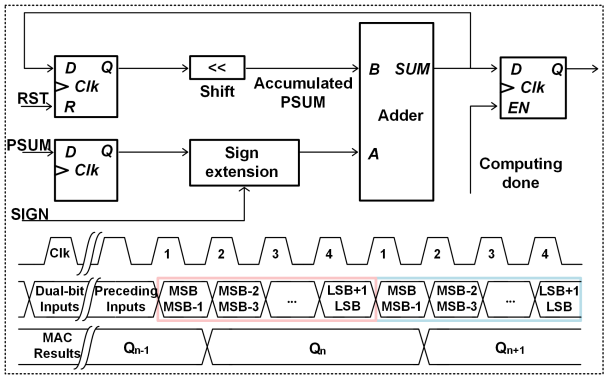


Fig. 6. The optimized accumulator and operating waveforms.

rates, computing their partial products separately can reduce energy consumption. For the input data's sign bit, the PSUM of the ATR will be taken signed negative first in advance and added to the shifted PSUM of the ATL to obtain PS_1. Finally, the accurate PSUM PS_2 is obtained by:

$$PS_2 = -P\sum w_i + (1-2P)PS_1 \tag{1}$$

where P is the value of the signal P. For the remaining bits, the PSUM of the ATR is added directly to the shifted PSUM of the ATL to obtain PS_1. The PS_2 is calculated by:

$$PS_2 = 3P\sum w_i + (1-2P)PS_1 \tag{2}$$

If P is 0, the input data is not inverted and the PSUM does not need to be corrected by the sum of the weights.

The inset of Fig. 5 shows the HSCU with the embedded hybrid computing unit. For INL, a 3T computing unit of transmission gate and pull-down NMOS is proposed to reduce energy consumption and area overhead without voltage loss. For INR, a NOR gate is utilized to ensure that it is sufficient to drive the shift logic circuits.

D. Optimized Accumulator

Fig.6 shows the optimized accumulator and operating waveforms. In the ITWP CIM macro operation, 256 dual-bit input data are fed into the macro in an MSB-first dual-bit-serial manner. In the first cycle, the MSB and MSB–1 are fed into the CIM array. Meanwhile, the accumulator performs a sign extension on the PSUM derived from the LSB and LSB+1 of the preceding inputs before adding it to the accumulated PSUM. Upon the next rising clock edge, the complete accumulation result of the preceding input set is output, and the intermediate accumulation state resets. Therefore, the proposed accumulator can work continuously without an additional cycle for outputting the result and resetting as in [9]. This continuous operation reduces the static energy consumption and increases the throughput of the CIM array.

III. EVALUATION RESULTS

In this work, a 32Kb ITWP CIM macro using a 28nm CMOS process is implemented. Fig. 7 shows the macro's layout with an area of 0.727 mm^2. Fig. 8(a) shows the performance of the macro compared to conventional bit-serial CIM [9]. The results show that the ITWP CIM macro has

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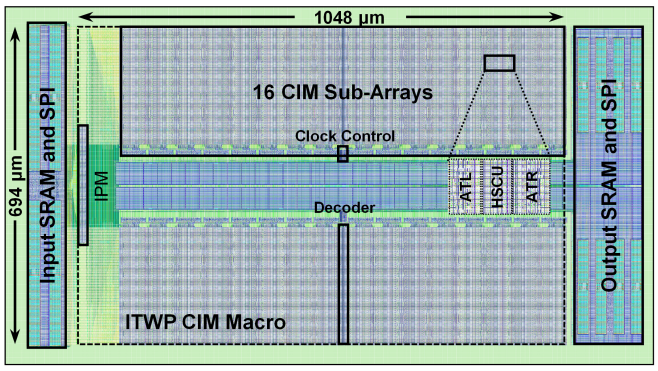


Fig. 7. The layout of the ITWP CIM macro.

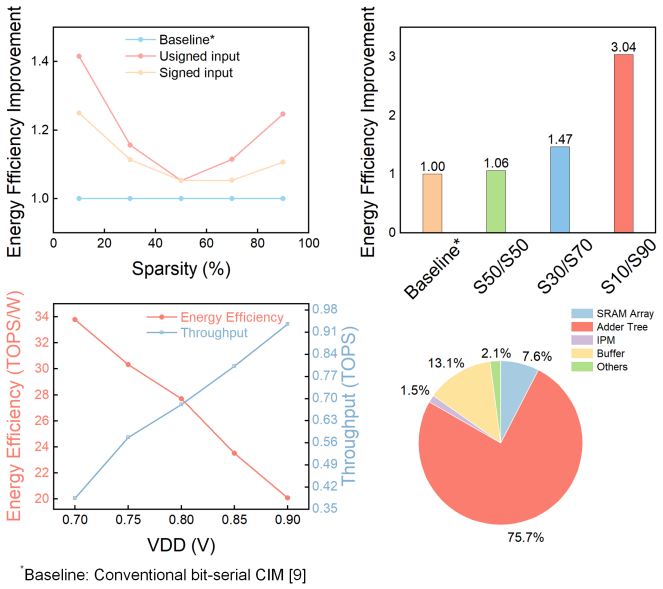


Fig. 8. (a) Energy efficiency improvement of the ITWP CIM macro when computing input data with various sparsity levels. (b) Energy efficiency improvement when sequentially processing input data with different sparsity levels. (c) Energy efficiency and throughput at different supply voltages. (d) Power breakdown of the ITWP CIM macro.

higher energy efficiency with input data at different sparsity levels. In particular, when the sparsity is less than 50%, the ITWP CIM achieves even more outstanding energy efficiency improvement due to input sparsification, which is improved up to 1.42x.

Fig. 8(b) shows the energy efficiency improvement of the ITWP CIM macro when sequentially processing input data with different sparsity levels. In particular, when sequentially processing input data with sparsity levels of 10% and 90%, the ITWP CIM macro shows a remarkable improvement, achieving up to 3.04x energy efficiency compared to conventional bit-serial computational CIM.

Fig. 8(c) shows the energy efficiency and throughput of the ITWP CIM macro at various supply voltages. The peak energy efficiency is 33.78 TOPS/W, which is achieved under a supply voltage of 0.7 V (signed 8b input/signed 8b weight, 10% input toggle rate, 90% input sparsity, 50% 1s for weights). Under a supply voltage of 0.9V, the peak throughput of 0.935 TOPS is achieved. Fig. 8(d) shows the power breakdown of the ITWP CIM macro.

TABLE III. COMPARISON TO PREVIOUS WORKS

Reference	ISSCC 2021 [9]	TCAS-I 2023 [6]	ISCAS 2024 [5]	TCAS-I 2024 [7]	This Work
Technology	22nm	65nm	28 nm	55nm	**28nm**
MAC approach	Digital	Analog	Digital	Digital	**Digital**
Array Size	64Kb	101Kb	17Kb	4Kb	**32Kb**
Macro Area (mm^2)	0.202	2.157	0.342	0.062	**0.727**
Voltage (V)	0.72	1	0.75 – 0.9	0.7 – 1.2	**0.7 – 0.9**
Input Precision (bit)	1 – 8	8	8	1 – 8	**1 – 8**
Weight Precision (bit)	4/8/12/16	8	8	1 – 8	**4/8**
Output Precision (bit)	16(4b/4b) 24(8b/8b)	8	19	7-22	**16(4b/4b) 24(8b/8b)**
Frequency (MHz)	500	240 – 1000	N.A.	100	**190 – 450**
8b Throughput (TOPS)	0.917	0.0358 – 0.0512	N.A.	0.0128	**0.385 – 0.935**
8b Area Eff. ($TOPS/mm^2$)	4.59	0.02	0.059	0.21	**1.23**
8b Energy Eff. (TOPS/W)	24.7 [2,4]	10.3 [2,4]	8.36 [1,4]	14.2 [2,3]	**20.08 – 33.78** [1,3,4]

[1] Post-layout simulation.
[2] Chip measurement.
[3] 10% input toggle rate.
[4] 50% 1s for weights.

IV. COMPARISON TO PREVIOUS WORKS

Table III compares the performance of the proposed ITWP CIM macro with some existing SRAM-based CIM macros. Due to the sparsification and the restructuring of the input data, the ITWP CIM macro achieves a better peak energy efficiency of 33.78 TOPS/W with signed 8b input and weight. In addition, the ITWP CIM macro achieves a peak throughput of 0.935 TOPS due to the optimized accumulation and dual-bit input parallel computing mode.

V. CONCLUSION

The energy consumption of SRAM-based digital CIM macros is significantly influenced by the input data's toggle rate and sparsity. Therefore, this work proposes an ITWP CIM macro that can invert low-sparsity input data and restructure dual-bit input data to increase the sparsity and reduce the toggle rate. The hybrid SRAM CIM unit and the split adder tree are proposed to simultaneously calculate the transformed dual-bit input data, resulting in higher energy efficiency and throughput. The errors caused by the input transformation are corrected by the precomputed sum of the weights with minimal overhead. In addition, an optimized uninterrupted accumulator is proposed to improve the throughput further. With a 28 nm process, the ITWP CIM macro can achieve 1.23 $TOPS/mm^2$ in area efficiency. When operating with signed 8b input and signed 8b weight, the ITWP CIM macro achieves a peak throughput of 0.935 TOPS and an energy efficiency of 20.08–33.78 TOPS/W.

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